How much is the wind worth? Wind effects on sprint and horizontal-jump performance across age and ability in a national competition registry

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Abstract

Every empirical estimate of how wind affects track-and-field performance comes from elite seniors, yet wind rules — the ±2.0 m/s legality threshold above all — apply identically to ten-year-olds and world-class sprinters. Using the Norwegian Athletics Federation's complete competition registry, which records gauge readings down to the youth 60 m, we estimated wind effects within athlete × season across five sprint and four horizontal-jump events at ages 10–34: 433 566 analysed results, of which 315 288, lying in 79 330 athlete-season-event cells with at least two wind-measured marks, identify the estimates, with month and round controls and standard errors clustered on meet. A +1 m/s tailwind was worth 0.059 s on 60 m, 0.082 s on 100 m, 0.121 s on 200 m, 2.9 cm in the long jump and 4.1 cm in the triple jump — roughly double the published elite estimates, which our senior stratum reproduces as the endpoint of a gradient. The absolute sprint effect declined with age (200 m, males: 0.143 s per m/s at 10–11 to 0.084 at 20–34) and with ability within age, scaling approximately with race duration: per second of race time the effect was 0.0059–0.0063 s per m/s in the straight sprints against 0.0040–0.0043 in the bend events. Headwinds cost more than tailwinds gave in sprints (100 m: 0.107 vs. 0.065 s per m/s, $p < 0.001$), an asymmetry larger than a v^2 drag benchmark predicts. In a post-plan extension, class-specific 60 m hurdles matched the flat 60 m, while the 100 and 110 m hurdles mostly exceeded the flat 100 m (by up to 48 %), partly because they take longer to run. Valued from the tailwind slope, the +2.0 m/s window was worth 0.14–0.17 s on the 100 m at 12–13 against 0.11–0.13 s among seniors — everywhere 0.2–0.7 of the athlete's own within-season standard deviation. Uniform wind rules thus permit the largest absolute wind contribution exactly where raw marks are most used to rank and select, and the registry already contains the age-specific corrections estimated here.

Keywords: wind; sprint; long jump; youth athletics; measurement; age-group competition; registry study

1 Introduction

Wind is the one weather variable athletics regulates. Sprint and horizontal-jump results count for records only when tailwind does not exceed 2.0 m/s, gauge readings are part of the official result, and seasonal lists in many countries separate legal from wind-aided marks. How much the wind actually moves a result has been estimated in two ways: physical and quasi-physical models of the sprint (Linthorne, 1994; Mureika, 2001), and data-driven analyses of elite competition results (Moinat et al., 2018; Hollings et al., 2012). Both traditions agree on the order of magnitude for elite seniors — roughly 0.04–0.06 s per m/s in the 100 m, somewhat more for women than for men — and both concern elite seniors only. Reviews of sprint testing likewise treat wind as a condition to standardise, with correction factors derived from adult data (Haugen and Buchheit, 2016).

Yet nothing guarantees that elite estimates transfer down the age and ability range, and the physics pulls in two directions. A slower runner spends longer on the course — a 12-year-old needs roughly a third more time over 100 m than a senior — so any given wind acts on them longer, inflating the absolute time effect. But aerodynamic drag scales with the square of airspeed, so slower runners spend a smaller share of their power against air, deflating it; children's higher surface-to-mass ratio pushes the other way again. The net scaling of the wind effect across age and ability is an empirical question that elite data cannot answer, and it matters practically: age-group records, seasonal youth lists, age-graded scoring tables and qualification standards all treat marks within ± 2.0 m/s as equivalent, in every age class, and talent evaluation leans on exactly these marks (Guttormsen, 2026).

The Norwegian registry is unusually well placed to answer it. Every sanctioned result in Norway is registered, and wind is measured and recorded down to the outdoor 60 m sprints that children contest from age 10 — to our knowledge the only population-scale source of wind-measured sprint data on children. We pre-specified five questions: the direction of the effect within athlete and season (H1, a design check); how the absolute effect scales with age (H2 — the exposure-time and aerodynamic-share mechanisms predict opposite directions); how the percentage effect scales (H3); whether headwinds cost more than tailwinds give (H4, as aerodynamics predicts); and what the ± 2.0 m/s window is worth in each class, expressed against

athletes' own performance variability (H5). The analysis plan was committed to version control before estimation; the study was not externally preregistered. At the author's initiative after the main estimations, class-specific short hurdles were added as a documented extension.

2 Methods

2.1 Data

We used a frozen extract (29 October 2026) of the Norwegian Athletics Federation's results database, which is electronically complete from 2011. The main analysis covers all results from 2011 onwards in nine wind-measured events: 60 m, 80 m, 100 m, 150 m and 200 m sprints, and long jump, long jump from a take-off zone, triple jump and triple jump from a zone (in the zone variants, used in the youngest classes, take-off is from a marked zone rather than a board, which reduces fouls). The hurdles extension covers the ten short-hurdle event codes with substantive wind data (60 m to 110 m hurdles; each code specifies the class's hurdle height, so a code identifies an age-class specification). Only approved, outdoor, electronically timed results of athletes with known sex and birth year were analysed, at calendar-year ages 10–34, with event-specific plausibility bounds and wind readings within ± 8 m/s. Of 670 803 extracted rows in the main events, 433 566 entered the analysis. The largest exclusions were indoor results — principally the indoor 60 m (86 727 rows), long jump (40 200) and 200 m (35 869) — and 18 692 outdoor rows without a wind value (overall, 4.9 % of approved outdoor results lack a reading). Missing wind is a completeness rather than a selection concern: it is concentrated in the smallest meets (9.1 % of rows in the smallest third of meets by size, against 1.5 % in the largest third) and its yearly share fluctuates between 2 and 7 %. Of the analysed rows, 315 288 lie in athlete-season-event cells with at least two wind-measured results and identify the estimates; the hurdles extension contributed 33 386 analysed rows. Times are analysed in seconds and jumps in metres (reported in centimetres); units were verified against known magnitudes per event.

2.2 Statistical model

The identifying variation is within athlete × season × event: the same athlete, in the same event and season, racing under different winds. For each event we estimated

$$y_{it} = \alpha_{i(\text{athlete}\times\text{year})} + \beta \cdot \text{wind}_{it} + \text{month dummies} + \text{round dummies} + \varepsilon_{it},$$

by within-demeaning; 79 330 athlete-season-event cells contribute at least two wind-measured results, and estimates are reported only where at least 300 results in 100 athlete-seasons contribute. Because the gauge reading is constant within a race, the race — and conservatively the meet — is the natural clustering level (Moulton, 1986); standard errors are therefore clustered on meet, with athlete clustering (the analysis plan's original choice, which gives intervals roughly half as wide in the large sprints and similar ones elsewhere) retained as robustness. The month dummies are pooled across years; since the year is fixed within an athlete-season cell, they are equivalent to year × month controls. β is reported as the *gain per +1 m/s of tailwind*, in seconds or centimetres, and in percent from a log-outcome specification. Age-band (10–11 through 20–34) and sex-specific models address H2/H3; ability tertiles address ability scaling, with ability defined as the athlete-season mean of *wind-adjusted* performance (each mark corrected to still air with the event-level slope), so that the ability measure itself is not contaminated by wind exposure, and the contributing athlete-seasons divided into equal thirds within event × sex × age band; separate slopes for head- and tailwind address H4; a quadratic term probes curvature. For H5, the advantage of a +2.0 m/s tailwind over still air is twice the tailwind-branch slope for the event, sex and age band — using the symmetric slope would overstate it, since headwinds cost more — expressed in seconds and as a share of the within-athlete-season variability of performance, estimated as the pooled within-cell standard deviation of wind-adjusted marks over cells with at least three results. Pre-specified robustness analyses required at least three results per cell, excluded readings of exactly 0.0, split the period (2011–2019 vs. 2021–2026; 2020, whose season was curtailed by the pandemic, belongs to neither half), and replaced month dummies with meet fixed effects (two-way demeaning); a post-plan check excluded readings within ± 0.2 of +2.0. Analyses used Python 3.13.7 with pandas 2.3.3, numpy 2.4.1 and statsmodels 0.14.6.

Two interpretive rules were fixed in advance. First, a single gauge reading does not measure the wind each athlete met; this measurement error attenuates estimates toward zero, so the sprint effects are read as lower bounds. The measurement protocol also differs by event family: under World Athletics rules the gauge beside the straight averages 10 s from the gun in the 100 m (13 s in the short hurdles), 10 s from when the leader enters the straight in the 200 m — so it measures the wind over only part of a course that is half bend — and 5 s per attempt in the jumps; Norwegian practice applies the same conventions to the domestic distances (60 and 80 m as the 100 m; 150 m as the 200 m). Comparisons across event families bear this in mind. Second, wind is not randomly assigned: identification relies on within-season comparisons with month, round and (in robustness) meet controls, and no causal language beyond this design sense is used. For the jumps, one further caveat applies: the registry stores the wind of the measured (best) attempt, and best attempts are more often wind-assisted, so jump estimates contain an attempt-selection component that pushes upward — the lower-bound reading applies to the sprints, while the jump estimates are better read as approximations of uncertain sign.

2.3 Ethics and transparency

The data are competition results registered by the Norwegian Athletics Federation: results from age 13 are publicly listed, while records for the 10–12 age classes are registered but not publicly released by the federation. No athlete is identified. Performance data below age 13 are used solely to estimate measurement conditions — not to rank, list or evaluate children — in keeping with the purpose of the Norwegian children's sport regulations; no result of an athlete under 13 appears in ranked, listed or identifiable form in the paper or its outputs. The project falls outside the Norwegian Health Research Act; an assessment from Sikt was obtained. The analysis plan was committed to version control before estimation; code and aggregate estimation files are available from the author, and athlete-level files are regenerable from the registry. Reporting follows STROBE where applicable (von Elm et al., 2007).

3 Results

3.1 Sample and wind exposure

Table 1 describes the nine main events. Wind readings centred near calm with wide spread (median +0.3 m/s; 5th–95th percentile −2.0 to +2.7). The distribution shows digit preference — 9.5 % of readings are exactly 0.0, more than any continuous wind distribution would place there and consistent with some unoperated gauges being recorded as calm, and 17 % fall on whole metres per second — and thins just above the legality line: 3.5 % of readings lie in (1.8, 2.0] against 2.1 % in (2.0, 2.2], with 1.7 % exactly +2.0. Both patterns motivate robustness checks: excluding exact zeros, and excluding readings within ±0.2 of +2.0, leave every estimate essentially unchanged (Table S2). The bulk of the material is young: the 60 m, contested outdoors by the youngest classes, is the largest single event (103 377 analysed results, of which 72 669 lie in athlete-seasons with at least two wind-measured races and thus contribute to estimation).

Table 1: Analysed events: volume, marks, and wind exposure (2011–2026).

Event	Results	In cells ≥2	Athlete-seasons	Athletes	Median mark	Wind, median (p5–p95)
60 m	103 377	72 669	19 374	10 799	9.59 s	+0.1 (−2.3–+2.5)
80 m	5 795	1 410	620	466	12.53 s	+0.0 (−2.6–+2.5)
100 m	76 598	64 756	13 253	5 468	12.88 s	+0.4 (−2.2–+2.8)
150 m	2 154	334	153	127	19.64 s	+0.3 (−2.5–+2.2)
200 m	76 261	59 416	15 571	6 621	27.99 s	+0.4 (−2.2–+2.9)
Long jump	85 789	51 870	12 204	6 404	4.05 m	+0.4 (−1.6–+2.6)
Long jump (zone)	54 665	44 336	12 338	7 835	3.64 m	+0.3 (−1.7–+2.7)
Triple jump	21 067	15 071	3 956	2 087	9.67 m	+0.6 (−1.3–+2.8)
Triple jump (zone)	7 860	5 426	1 861	1 413	8.42 m	+0.4 (−1.5–+2.8)

Results: analysed rows (approved, outdoor, electronic, ages 10–34, wind within ±8 m/s). In cells ≥2: the estimation sample (the n of Table 2). Athlete-seasons and athletes: units contributing to estimation in that event. Counts and wind summaries are computed from the analysis file; the estimation files are available from the author.

3.2 What a metre per second is worth (H1)

Table 2 reports the main within-athlete estimates. A +1 m/s tailwind was worth 0.059 s (95 % CI 0.057–0.061) on 60 m, 0.082 s (0.080–0.085) on 100 m, 0.121 s (0.114–0.129) on 200 m, 2.9 cm (2.7–3.1) in the long jump and 4.1 cm (3.6–4.6) in the triple jump; in percentage terms, 0.40–0.65 % per m/s. All directions matched H1. The small 80 m and 150 m events gave consistent estimates with wide intervals.

Table 2: Gain per +1 m/s of tailwind, within athlete × season (95 % CIs clustered on meet).

Event	Gain per +1 m/s (95 % CI)	Percent per +1 m/s (95 % CI)	n	Athlete-seasons
60 m	0.059 s (0.057–0.061)	0.625 (0.605–0.645)	72 669	19 374
80 m	0.075 s (0.065–0.084)	0.630 (0.556–0.704)	1 410	620
100 m	0.082 s (0.080–0.085)	0.646 (0.627–0.664)	64 756	13 253
150 m	0.081 s (0.055–0.108)	0.467 (0.329–0.605)	334	153
200 m	0.121 s (0.114–0.129)	0.442 (0.415–0.469)	59 416	15 571
Long jump	2.88 cm (2.67–3.10)	0.641 (0.590–0.692)	51 870	12 204
Long jump (zone)	2.11 cm (1.94–2.29)	0.570 (0.521–0.619)	44 336	12 338
Triple jump	4.10 cm (3.60–4.60)	0.398 (0.349–0.448)	15 071	3 956
Triple jump (zone)	3.88 cm (3.06–4.70)	0.449 (0.353–0.545)	5 426	1 861

Point estimates are identical under athlete clustering (the analysis plan's original choice), whose intervals are roughly half as wide in the large sprints and similar elsewhere; both sets are in the estimation files available from the author.

These population-level effects are roughly double the published elite estimates. That is not a contradiction but a continuum: restricted to seniors (20–34), our 100 m estimate falls to 0.071 s per m/s for men and 0.072 for women — and our senior men, averaging about 11.4 s on the 100 m, are themselves slower than the elite samples behind the published figures of roughly 0.04–0.06 s per m/s (Linthorne, 1994; Moinat et al., 2018). The elite literature, it turns out, has been describing the fast end of a gradient.

3.3 The effect falls with age — and with speed (H2, H3)

Figure 1 and Table 3 show the core result. In absolute seconds, the wind effect declined with age in every sprint and both sexes — monotonically in every series except the sparse girls' 200

m at 10–11: on 200 m for boys and men from 0.143 s per m/s at 10–11 to 0.084 at 20–34; on 60 m from 0.070 at 10–11 to 0.041 at 16–17; on 100 m from 0.101 at 12–13 to 0.071 at 20–34. The exposure-time mechanism wins: mean race durations shrink in almost the same proportion (men's 100 m, 14.96 s at 12–13 vs. 11.42 s at 20–34, a ratio of 1.31, against a wind-effect ratio of 1.42), so the percentage effect (Figure 2) is far flatter — but not constant, declining for instance from 0.70 to 0.62 % per m/s on the men's 100 m and from 0.70 to 0.52 % on the 60 m. Dividing each event's effect by its mean race duration makes the scaling explicit (Table S4): the straight sprints line up at 0.0059–0.0063 s per m/s per second of race time, while the bend events sit a third lower (150 m 0.0040, 200 m 0.0043) — as expected when the gauge, read beside the home straight, measures the wind over only part of a course that is half bend, and when the wind meets the runner at varying angles through the bend. In the jumps the absolute effect runs the other way, rising with age in centimetres (long jump, girls/women: 1.9 cm per m/s at 12–13 to 4.1 at 20–34; boys/men flatter) as jump distances and flight times grow, while the percentage effect shows no consistent age direction (Table S1) — the same exposure logic, with exposure now flight rather than time on the track. Within age bands, ability scaling follows the sprint pattern (Figure 3): with ability measured as the wind-adjusted season mean, the lowest tertile gained most per m/s in every sprint (100 m: 0.088 s against 0.079 in the highest tertile; 60 m: 0.066 against 0.056; 200 m: 0.132 against 0.117), while the jump tertiles did not separate. Slower athletes — whether young or simply weaker — are exposed longer, and no single correction factor, absolute or proportional, is exactly age-invariant.

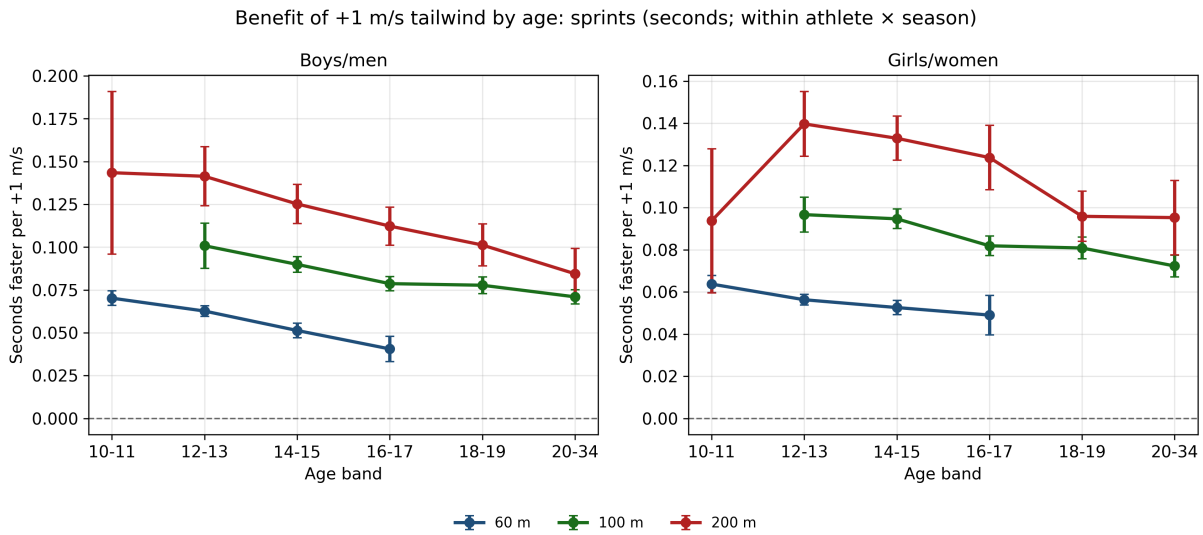


Figure 1: Benefit of +1 m/s tailwind by age band, in seconds, for sprints (within athlete × season; 95 % CIs clustered on meet).

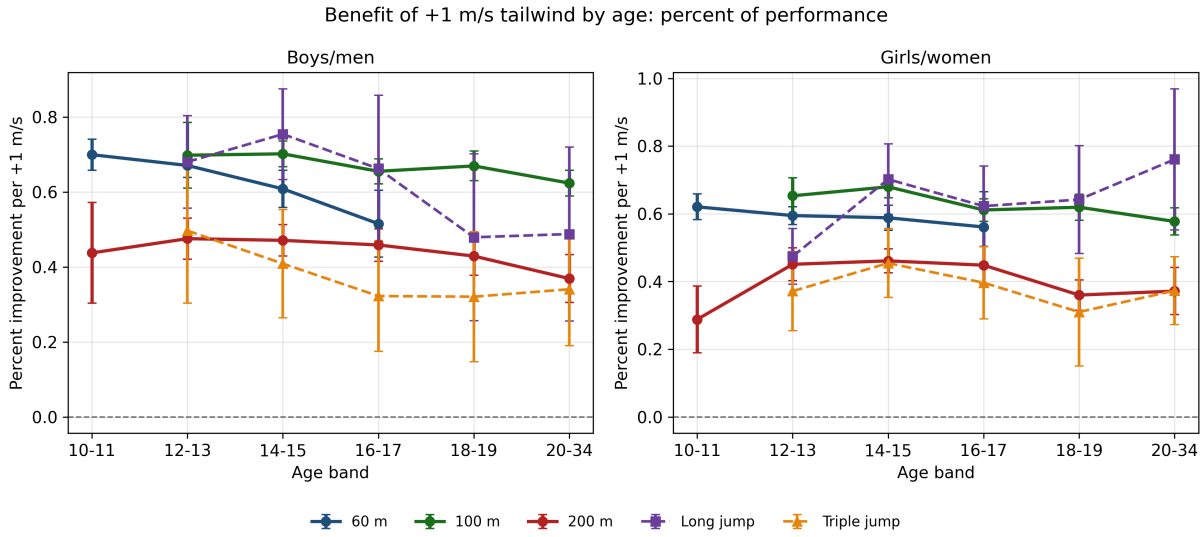


Figure 2: The same scaling in percent of performance, for sprints (solid lines) and horizontal jumps (dashed). For the sprints the percentage effect is far flatter across age than the absolute effect, but still declines; the jumps show no consistent age direction (95 % CIs clustered on meet).

Table 3: Gain per +1 m/s of tailwind (seconds, 95 % CI clustered on meet) by age band and sex, sprints.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.070 (0.066–0.075)	0.063 (0.060–0.066)	0.051 (0.047–0.056)	0.041 (0.033–0.048)	—	—
60 m	Girls/women	0.064 (0.060–0.068)	0.056 (0.054–0.059)	0.053 (0.049–0.056)	0.049 (0.040–0.058)	—	—
100 m	Boys/men	—	0.101 (0.088–0.114)	0.090 (0.085–0.094)	0.079 (0.074–0.083)	0.078 (0.073–0.083)	0.071 (0.067–0.075)
100 m	Girls/women	—	0.097 (0.088–0.105)	0.095 (0.090–0.099)	0.082 (0.077–0.086)	0.081 (0.076–0.086)	0.072 (0.067–0.077)
200 m	Boys/men	0.143 (0.096–0.191)	0.141 (0.124–0.159)	0.125 (0.114–0.137)	0.112 (0.101–0.123)	0.101 (0.089–0.114)	0.084 (0.070–0.099)
200 m	Girls/women	0.094 (0.059–0.128)	0.140 (0.124–0.155)	0.133 (0.122–0.143)	0.124 (0.108–0.139)	0.096 (0.084–0.108)	0.095 (0.078–0.113)

Dashes: fewer than 300 results in 100 athlete-seasons, the reporting threshold (60 m above 17, which is rarely run outdoors; 100 m at 10–11, where 60 m is the standard sprint). Table values are reproduced from the estimation files available from the author.

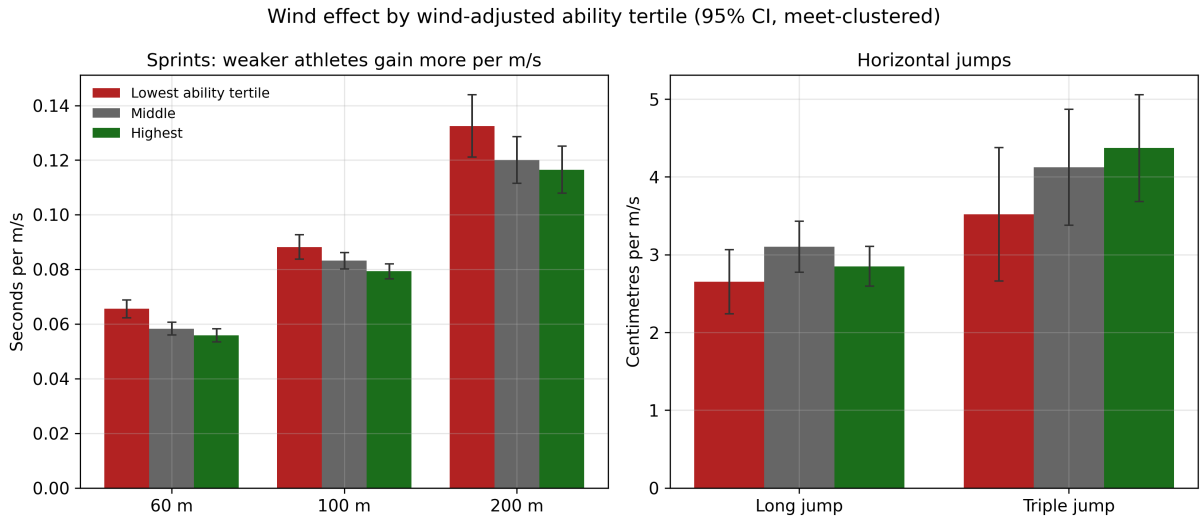


Figure 3: Wind effect by ability tertile: athlete-seasons divided into equal thirds of the wind-adjusted season mean within event × sex × age band (95 % CIs clustered on meet).

3.4 Headwinds cost more than tailwinds give (H4)

In every large sprint event the headwind slope exceeded the tailwind slope (Figure 4; Table 4): on 100 m a -1 m/s headwind cost 0.107 s against a +1 m/s tailwind's 0.065 s ($p < 0.001$ for equal slopes); on 200 m, 0.184 against 0.078; on 60 m, 0.072 against 0.047. The positive quadratic terms tell the same story (200 m: +0.0141 s per $(\text{m/s})^2$, SE 0.0011; Table S5). The direction matches drag physics, but the size does not: with drag proportional to the square of airspeed, the cost/gain ratio for a ± 1 m/s wind at these running speeds is $(2v + 1)/(2v - 1) \approx$

1.14–1.18, while the observed ratios are 1.51 (60 m), 1.66 (100 m) and 2.37 (200 m). The excess is consistent with the lower-bound logic — gauge error attenuates both slopes, but selective behaviour (racing, wind-waiting) plausibly dampens the tailwind side more — and on the 200 m, where the excess is largest, the gauge reads only the home straight; we therefore read the asymmetry as directional, not as a calibrated magnitude. In the jumps the asymmetry was absent (long jump, p = 0.999) or reversed in the zone and triple-jump variants — a pattern consistent with attempt selection, since the recorded wind belongs to the best attempt, and best attempts are more often wind-assisted.

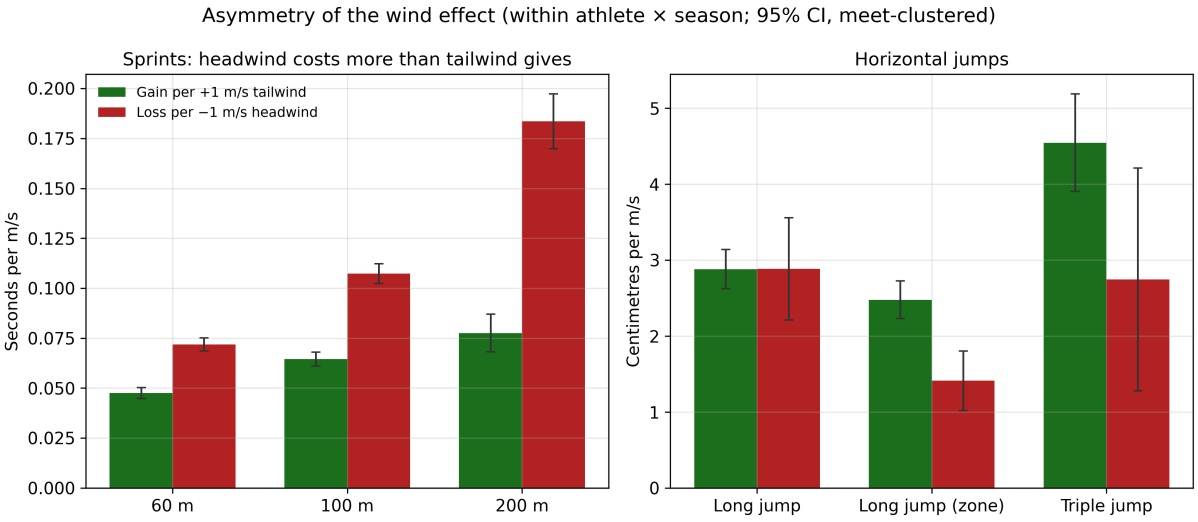


Figure 4: Separate head- and tailwind slopes (within athlete × season; 95 % CIs clustered on meet).

Table 4: Asymmetry: separate tail- and headwind slopes, observed cost/gain ratio, and the v^2 drag benchmark (95 % CIs clustered on meet).

Event	Unit	Tailwind gain per +1 m/s	Headwind loss per −1 m/s	Ratio	v^2 benchmark	p equal slopes
60 m	s	0.047 (0.045–0.050)	0.072 (0.069–0.075)	1.51	1.18	<0.001
80 m	s	0.072 (0.054–0.090)	0.078 (0.060–0.095)	1.07	1.17	0.741
100 m	s	0.065 (0.061–0.068)	0.107 (0.102–0.112)	1.66	1.14	<0.001
150 m	s	0.093 (0.048–0.137)	0.067 (−0.004–0.138)	0.72	1.14	0.623
200 m	s	0.078 (0.068–0.087)	0.184 (0.170–0.197)	2.37	1.15	<0.001
Long jump	cm	2.88 (2.63–3.14)	2.88 (2.21–3.56)	1.00	—	0.999
Long jump (zone)	cm	2.48 (2.23–2.73)	1.41 (1.02–1.80)	0.57	—	<0.001
Triple jump	cm	4.55 (3.91–5.19)	2.74 (1.28–4.21)	0.60	—	0.048
Triple jump (zone)	cm	4.69 (3.65–5.74)	1.90 (0.16–3.64)	0.40	—	0.013

The benchmark is $(2v + 1)/(2v - 1)$ at the event's mean speed v, the cost/gain ratio of ± 1 m/s under drag proportional to the square of airspeed. No benchmark is given for the jumps, where wind acts on run-up and flight differently.

3.5 Short hurdles: like flat sprints over 60 m, larger over 100 and 110 m

In the age-class system, hurdle events carry class-specific heights, so each event code identifies an age-class specification. Across the ten codes with wind data (Table 5; Figure 5), the 60 m hurdles matched the flat 60 m of the same sex and class: 0.055–0.067 s per m/s against 0.056–0.070, point estimates from 21 % below to 9 % above the flat references, with overlapping intervals. Over 100 and 110 m the hurdle estimates mostly exceeded the flat 100 m reference, from 4 % below to 48 % above, and in two youth classes — girls' 76.2 cm (0.108 against 0.082) and boys' 84 cm (0.133 against 0.090) — the hurdle interval lies entirely above the flat reference. Per second of race time the pooled hurdle codes (0.0048–0.0069 s per m/s) span the straight-sprint rates (Table S4), so the hurdles' longer race times account for much of the difference, though not all of it in those two youth classes. The age gradient reappeared within a distance: on the 100 m hurdles, 0.108 s per m/s for girls' 76.2 cm at 16–17 and 0.133 for boys' 84 cm at 14–15, against 0.069 for the senior women's 84 cm event.

Table 5: Wind effect in short hurdles by class-specific event code (sex-specific rows; flat-sprint reference for the same sex and distance class in that sex's modal age band for the code; 95 % CIs clustered on meet).

Event (hurdle height)	Sex	Modal band	Gain per +1 m/s (95 % CI), s	Flat reference (95 % CI), s	n
60 m hurdles 68 cm	Boys	10–11	0.055 (0.037–0.073)	0.070 (0.066–0.075)	1 215
60 m hurdles 68 cm	Girls	10–11	0.067 (0.051–0.082)	0.064 (0.060–0.068)	2 562
60 m hurdles 76.2 cm	Boys	12–13	0.063 (0.050–0.075)	0.063 (0.060–0.066)	3 457
60 m hurdles 76.2 cm	Girls	12–13	0.062 (0.052–0.071)	0.056 (0.054–0.059)	7 293
80 m hurdles 76.2 cm	Girls	14–15	0.072 (0.054–0.089)	—	2 529
80 m hurdles 84 cm	Boys	14–15	0.078 (0.055–0.100)	—	1 474
100 m hurdles 76.2 cm	Girls	16–17	0.108 (0.091–0.125)	0.082 (0.077–0.086)	1 055
100 m hurdles 84 cm	Boys	14–15	0.133 (0.097–0.168)	0.090 (0.085–0.094)	709
100 m hurdles 84 cm	Women	20–34	0.069 (0.043–0.095)	0.072 (0.067–0.077)	2 882
100 m hurdles 91.4 cm	Boys	16–17	0.084 (0.057–0.111)	0.079 (0.074–0.083)	685
110 m hurdles 91.4 cm	Boys	16–17	0.103 (0.044–0.162)	0.079 (0.074–0.083)	583
110 m hurdles 100 cm	Men	18–19	0.096 (0.044–0.149)	0.078 (0.073–0.083)	657
110 m hurdles 106.7 cm	Men	20–34	0.085 (0.051–0.119)	0.071 (0.067–0.075)	623

The hurdles analysis was added after the main estimations at the author's initiative and is documented as a post-plan extension. Flat references are the 60 m or 100 m estimate for the same sex in that sex's modal age band; the flat 80 m has too few results at 14–15 for a reference. Codes with fewer than 300 results for one sex have only the other sex's row.

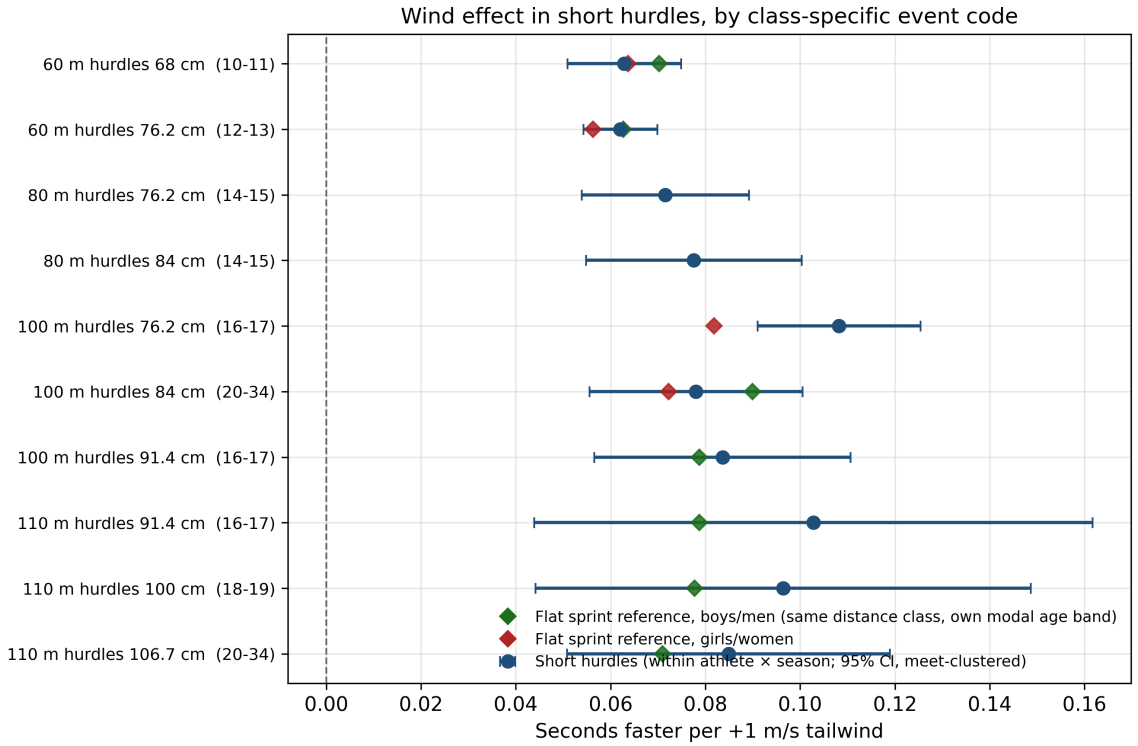


Figure 5: Short hurdles: wind effect per class-specific code (pooled sexes; dots with 95 % CIs clustered on meet) against flat-sprint references for each sex (diamonds; boys/men green, girls/women red).

3.6 What the legal window is worth in each class (H5)

Table 6 translates the estimates into the quantity the rules regulate: the advantage of a +2.0 m/s tailwind over still air, valued from the tailwind-branch slope (Section 2.2) and expressed in seconds and as a share of the athlete's own within-season variability. In seconds, the window is worth most where athletes are slowest: 0.14 s on the boys' 100 m at 12–13 and 0.17 s for girls, against 0.13 and 0.11 s among seniors; 0.21 s on the boys' 200 m at 10–11 against 0.10 s for senior men. Expressed against the athlete's own variability, the window is worth 0.2–0.7 SD across events, sexes and ages, with no systematic youth excess: young athletes' marks scatter more for many reasons, so the larger seconds do not translate into a larger share of their own variation. Two external yardsticks complete the picture. On the federation's age-graded scoring tables, +2.0 m/s on the 100 m converts to 24–25 points at age 13 against 19–22 points on the oldest (age-19) table — similar, because the senior tables are steeper per second. And in the

2025 national 100 m season lists, correcting tailwind-aided marks to still air with the class-specific slopes changes the athlete occupying 4–5 of the top-10 places in the class-14 lists and 9 of 10 in the senior men's list, where 86 % of top-50 marks carried tailwinds above +1 m/s: wind moves rankings at every age, and most where lists are decided by hundredths.

Table 6: Advantage of a +2.0 m/s tailwind over still air, valued from the tailwind-branch slope: seconds, and share of the within-athlete-season SD of wind-adjusted performance.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.11 s (0.51 SD)	0.11 s (0.55 SD)	0.09 s (0.59 SD)	0.04 s (0.35 SD)	—	—
60 m	Girls/women	0.09 s (0.43 SD)	0.09 s (0.52 SD)	0.08 s (0.49 SD)	0.04 s (0.37 SD)	—	—
100 m	Boys/men	—	0.14 s (0.36 SD)	0.14 s (0.66 SD)	0.12 s (0.65 SD)	0.13 s (0.39 SD)	0.13 s (0.50 SD)
100 m	Girls/women	—	0.17 s (0.70 SD)	0.14 s (0.65 SD)	0.12 s (0.63 SD)	0.13 s (0.73 SD)	0.11 s (0.67 SD)
200 m	Boys/men	0.21 s (0.24 SD)	0.15 s (0.21 SD)	0.16 s (0.30 SD)	0.15 s (0.28 SD)	0.14 s (0.31 SD)	0.10 s (0.23 SD)
200 m	Girls/women	0.20 s (0.21 SD)	0.21 s (0.32 SD)	0.19 s (0.36 SD)	0.13 s (0.28 SD)	0.13 s (0.37 SD)	0.09 s (0.29 SD)

Seconds are 2× the tailwind-branch slope (Table S3 gives the slopes with CIs). SD shares use the pooled within-cell standard deviation of wind-adjusted marks over cells with at least three results, per event × sex × band — a denominator free of the wind-driven variance itself and of singleton-cell bias. Values reproduced from the estimation files available from the author. The 60 m rows at 16–17 rest on thin outdoor samples.

3.7 Robustness

Estimates were essentially unchanged when cells required three or more results, when exact-zero readings were excluded, when readings within ±0.2 of +2.0 were excluded, across period halves, and under athlete instead of meet clustering (Table S2; point estimates are identical under either clustering). Replacing month dummies with meet fixed effects attenuated the 60, 100 and 200 m estimates by 13–41 % (100 m: 0.082 to 0.066 s per m/s), left the small 80 and 150 m events essentially unchanged, and raised the jump estimates by 26–45 %. In the sprints the attenuation is the expected arithmetic of measurement error: only 25–35 % of their wind variance lies within meets (40–53 % in the jumps, where attempts spread across the day), so meet effects absorb most of the signal and a fixed gauge error looms correspondingly larger; under classical race-level error, a gauge-error SD of 0.40–0.58 m/s (60 to 200 m) would reproduce the observed drops, with reliability falling from 0.85–0.92 to 0.50–0.80. The equivalent reading — that the month-dummy estimates carry some between-meet confounding — cannot be excluded, and the rise in the jumps, which measurement error alone would not produce, is harder to read given

attempt selection. We therefore read the primary and meet-FE specifications as bracketing the plausible range in the sprints, both of them lower bounds under wind measurement error.

4 Discussion

4.1 Principal findings

In the first population-scale estimates of wind effects across age and ability, a metre per second of tailwind was worth 42–70 % more, in absolute terms, to a boy sprinter than to a senior man — 0.10 versus 0.07 s on 100 m at 12–13, 0.14 versus 0.08 s on 200 m at 10–11 — and 34–47 % more among girls aged 12–13, with the published elite values (Linthorne, 1994; Mureika, 2001; Moinat et al., 2018; Hollings et al., 2012) emerging as the fast endpoint of a continuous gradient. The gradient's dominant mechanism is exposure time: effects scaled approximately with race duration, across age and equally across ability within age, and the percentage effect was accordingly much flatter — though still mildly declining. The residual decline cannot be the aerodynamic share, which grows with speed and would push the percentage effect the other way; it is consistent with a fixed wind being a larger fraction of a slower runner's airspeed, and perhaps with children's higher surface-to-mass ratio. Headwinds cost sprinters more than tailwinds gave them — directionally as drag physics predicts and as the positive curvature confirms, though more asymmetrically than the v^2 benchmark, so we treat the direction, not the magnitude, as established; in a post-plan extension, class-specific short hurdles matched flat sprints over 60 m and exceeded them over 100 and 110 m, partly through longer race times.

4.2 Implications

For measurement, the results supply what has been missing: empirically grounded, age- and event-specific wind values for youth athletics, with a full correction table by event, sex and age band in Table S3. Since neither the absolute nor the percentage effect is strictly invariant, corrections should be age-graded; a single senior-derived factor understates youth wind effects by up to two-fifths. For fairness, the $\pm$2.0 m/s legality window is not age-neutral in the units lists are kept in: it licenses the largest contribution in seconds, and the largest share of race

time, in the youngest and slowest classes. Relative to athletes' own within-season variability the window is worth a broadly similar 0.2–0.7 SD at every age, and on the age-graded scoring tables similar points — so the inequity is specific to raw-seconds comparisons, which is exactly what season lists, records and qualifying standards are. The 2025 list exercise makes it concrete: still-air correction reshuffles top-10 places at every age, senior lists included. The practical conclusions are modest and do not require rule changes: report and store wind with youth results (as Norway does), prefer wind-adjusted or wind-stratified comparisons when marks inform selection or records, and treat single wind-assisted breakthroughs with proportionate scepticism. For talent evaluation the point compounds a broader lesson of the identification literature — early performance is an unreliable selection signal (Vaeyens et al., 2008; Guttormsen, 2026): at the ages where selection decisions lean hardest on raw times, those times carry the most weather per second of racing.

4.3 Strengths and limitations

Strengths include scale (433 566 analysed results, 315 288 of them identifying; 79 330 athlete-season-event cells), a within-athlete design that removes stable athlete and seasonal composition, direction-and-magnitude agreement with the elite literature at the senior endpoint, internal consistency across age, ability, events and the hurdles extension, and a plan committed before estimation. Limitations are stated as read rules: gauge readings mismeasure athlete-experienced wind, so the sprint estimates are lower bounds, and the meet-FE attenuation quantifies how much identification leans on between-race wind variation; the gauge protocol differs by event family, and for the bend events measures only part of the course, so cross-family comparisons (and the low per-second rates of the 150/200 m) partly reflect measurement rather than physiology; wind is not randomised, and within-season form-weather covariation is controlled only by month, round and meet terms; jump estimates carry an upward attempt-selection component because wind attaches to the measured attempt, so the lower-bound reading does not extend to them; the senior stratum is 20–34, so masters athletics is outside scope; missing wind (4.9 % of approved outdoor results, concentrated in the smallest meets) limits completeness but gives no mechanism for bias within athlete-seasons; temperature and other weather are unobserved

(a meteorological linkage is a natural extension); and the setting is one federation's registry, though the mechanisms are physical and should travel.

5 Conclusion

Wind effects in athletics are not a constant to be borrowed from elite studies. They are 34–70 % larger for child sprinters than for seniors because slower athletes are exposed longer, they fall with ability for the same reason, and they are asymmetric in the direction drag physics predicts. The $\pm$2.0 m/s window was calibrated on the athletes for whom a metre per second costs the fewest seconds, and it licenses the largest raw-seconds wind contribution in the classes that race slowest. The remedy requires no new rules, because it operates on lists rather than races: wind is already recorded with every result, and the age-specific corrections estimated here (Table S3) can make lists, records and talent evaluation measurably fairer — for seniors too — while every race is run exactly as before.

Declarations

Funding. The author received no funding for this work.

Competing interests. The author declares no competing interests. The author maintains an open, non-commercial statistics platform built on the federation's public results database.

Ethics. Competition results registered by the Norwegian Athletics Federation (publicly listed from age 13; registered but not publicly released for the 10–12 age classes); no individual is identified; performance data below age 13 are used only to estimate measurement conditions, never to rank or evaluate children. Outside the scope of the Norwegian Health Research Act; Sikt assessment obtained.

Data availability. The underlying data are the Norwegian Athletics Federation's competition results; records for the 10–12 age classes are not publicly released and cannot be shared at the individual level. Analysis code and aggregate estimation files are available from the author; athlete-level files are regenerable from the registry.

Author contributions. AGG conceived the study, performed all analyses, and wrote the manuscript.

Declaration of AI use. The author used a large language model (Claude, Anthropic) for programming assistance and manuscript drafting support. The author reviewed, verified, and edited all content and takes full responsibility for the integrity of the work.

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Supplementary material

Table S1: Wind effect in the horizontal jumps by age band (gain per +1 m/s; values girls/women / boys/men; 95 % CIs clustered on meet).

Event	Unit	10–11	12–13	14–15	16–17	18–19	20–34
Long jump	cm	—	1.90 (1.58–2.22) / 2.69 (2.19–3.19)	2.90 (2.50–3.31) / 3.35 (2.88–3.81)	3.04 (2.54–3.54) / 3.70 (2.79–4.61)	3.20 (2.48–3.91) / 2.95 (1.75–4.15)	4.13 (3.13–5.13) / 3.08 (1.81–4.35)
Long jump	%	—	0.47 (0.39–0.56) / 0.68 (0.56–0.80)	0.70 (0.60–0.81) / 0.75 (0.63–0.88)	0.62 (0.50–0.74) / 0.66 (0.47–0.86)	0.64 (0.48–0.80) / 0.48 (0.26–0.70)	0.76 (0.55–0.97) / 0.49 (0.26–0.72)
Long jump (zone)	cm	1.83 (1.48–2.18) / 2.39 (1.96–2.81)	1.86 (1.63–2.10) / 2.54 (2.23–2.85)	—	—	—	—
Long jump (zone)	%	0.55 (0.44–0.66) / 0.70 (0.57–0.82)	0.49 (0.42–0.55) / 0.66 (0.58–0.74)	—	—	—	—
Triple jump	cm	—	3.31 (2.30–4.31) / 4.71 (2.89–6.54)	4.21 (3.27–5.15) / 4.19 (2.76–5.63)	4.13 (3.03–5.23) / 4.12 (2.37–5.87)	3.52 (1.85–5.18) / 4.21 (2.00–6.43)	4.43 (3.26–5.60) / 4.89 (2.86–6.93)
Triple jump	%	—	0.37 (0.25–0.49) / 0.50 (0.30–0.69)	0.45 (0.35–0.56) / 0.41 (0.26–0.55)	0.40 (0.29–0.50) / 0.32 (0.17–0.47)	0.31 (0.15–0.47) / 0.32 (0.15–0.49)	0.37 (0.27–0.47) / 0.34 (0.19–0.49)

In each cell the first value is girls/women, the second boys/men. The zone events are contested only in the youngest classes. In centimetres the effect rises with age for girls and women as jump distances and flight times grow; in percent it shows no consistent age direction.

Table S2: Robustness: gain per +1 m/s under alternative specifications (point estimates; s for sprints, cm for jumps).

Event	Primary	Cells $\geq$3	No exact 0.0	Excl. $\pm$0.2 of +2.0	2011–19	2021–26	Meet FE
60 m	0.0590	0.0589	0.0590	0.0587	0.0599	0.0573	0.0512
80 m	0.0746	0.0711	0.0761	0.0743	0.0740	0.0779	0.0769
100 m	0.0823	0.0816	0.0823	0.0827	0.0842	0.0795	0.0659
150 m	0.0814	—	0.0780	0.0865	—	—	0.0838
200 m	0.1214	0.1204	0.1214	0.1207	0.1258	0.1134	0.0712
Long jump	2.8827	2.8292	2.9551	2.9635	3.0308	2.7158	3.6465
Long jump (zone)	2.1133	2.0471	2.1538	2.1406	2.1337	2.1518	2.6621
Triple jump	4.0989	4.0130	4.1723	4.1485	3.8661	4.7803	5.9366
Triple jump (zone)	3.8821	3.3154	3.8589	3.7672	3.7075	4.8247	5.0965

Point estimates are identical under athlete and meet clustering; only intervals differ. The 150 m lacks the cells-$\geq$3 and period columns for want of data.

Table S3: Correction table: separate tail- and headwind slopes by event, sex and age band (sprints; jumps event-level, in cm), with 95 % CIs clustered on meet.

Event	Sex	Age band	Tailwind slope per +1 m/s (95 % CI)	Headwind slope per −1 m/s (95 % CI)	n
60 m	Boys/men	10–11	0.055 (0.048–0.062)	0.088 (0.079–0.097)	7 635
60 m	Boys/men	12–13	0.056 (0.051–0.061)	0.070 (0.064–0.077)	15 480
60 m	Boys/men	14–15	0.043 (0.037–0.048)	0.061 (0.052–0.071)	6 966
60 m	Boys/men	16–17	0.019 (0.007–0.031)	0.064 (0.053–0.076)	572
60 m	Girls/women	10–11	0.044 (0.038–0.051)	0.085 (0.078–0.092)	9 794
60 m	Girls/women	12–13	0.046 (0.042–0.051)	0.067 (0.062–0.072)	21 934
60 m	Girls/women	14–15	0.038 (0.033–0.043)	0.068 (0.063–0.074)	9 343
60 m	Girls/women	16–17	0.021 (0.010–0.031)	0.091 (0.074–0.108)	389
100 m	Boys/men	12–13	0.071 (0.048–0.095)	0.140 (0.117–0.164)	1 773
100 m	Boys/men	14–15	0.072 (0.065–0.080)	0.113 (0.104–0.122)	7 526
100 m	Boys/men	16–17	0.062 (0.055–0.069)	0.103 (0.095–0.110)	9 904
100 m	Boys/men	18–19	0.067 (0.058–0.076)	0.093 (0.081–0.106)	6 039
100 m	Boys/men	20–34	0.065 (0.058–0.071)	0.082 (0.072–0.092)	8 937
100 m	Girls/women	12–13	0.083 (0.066–0.100)	0.112 (0.095–0.129)	2 767
100 m	Girls/women	14–15	0.071 (0.064–0.078)	0.126 (0.117–0.134)	9 770
100 m	Girls/women	16–17	0.059 (0.053–0.065)	0.112 (0.103–0.121)	9 073
100 m	Girls/women	18–19	0.064 (0.057–0.071)	0.106 (0.094–0.119)	4 213
100 m	Girls/women	20–34	0.057 (0.051–0.064)	0.098 (0.088–0.108)	4 554
200 m	Boys/men	10–11	0.106 (0.039–0.172)	0.194 (0.112–0.276)	1 302
200 m	Boys/men	12–13	0.076 (0.047–0.104)	0.217 (0.182–0.252)	5 467
200 m	Boys/men	14–15	0.078 (0.062–0.093)	0.187 (0.167–0.208)	7 187
200 m	Boys/men	16–17	0.074 (0.062–0.087)	0.172 (0.149–0.195)	6 381
200 m	Boys/men	18–19	0.068 (0.055–0.082)	0.149 (0.126–0.171)	4 163
200 m	Boys/men	20–34	0.048 (0.028–0.067)	0.151 (0.128–0.175)	4 939
200 m	Girls/women	10–11	0.099 (0.040–0.157)	0.086 (0.008–0.164)	1 780
200 m	Girls/women	12–13	0.105 (0.084–0.125)	0.181 (0.149–0.212)	7 738
200 m	Girls/women	14–15	0.094 (0.078–0.110)	0.186 (0.166–0.207)	8 682
200 m	Girls/women	16–17	0.066 (0.051–0.081)	0.208 (0.178–0.239)	5 861
200 m	Girls/women	18–19	0.065 (0.049–0.080)	0.151 (0.123–0.179)	2 958
200 m	Girls/women	20–34	0.047 (0.027–0.067)	0.181 (0.148–0.214)	2 958
Long jump	Both	all	2.88 (2.63–3.14) cm	2.88 (2.21–3.56) cm	51 870
Long jump (zone)	Both	all	2.48 (2.23–2.73) cm	1.41 (1.02–1.80) cm	44 336
Triple jump	Both	all	4.55 (3.91–5.19) cm	2.74 (1.28–4.21) cm	15 071
Triple jump (zone)	Both	all	4.69 (3.65–5.74) cm	1.90 (0.16–3.64) cm	5 426

Worked example: a girls' 12–13 100 m run in +1.8 m/s is corrected by $1.8 \times 0.083 = 0.15$ s to still air. Sprint slopes are in seconds; the four jump rows are in centimetres, pooled across bands. The 60 m rows at 16–17 rest on thin outdoor samples.

Table S4: Effect per second of race time: gain per +1 m/s divided by mean race duration, flat sprints and short hurdles.

Event	Mean time (s)	Gain per +1 m/s (s)	Per second of race time
60 m	9.67	0.059	0.00610
80 m	12.58	0.075	0.00594
100 m	13.08	0.082	0.00629
150 m	20.22	0.081	0.00403
200 m	28.47	0.121	0.00427
60 m hurdles 68 cm	13.16	0.063	0.00478
60 m hurdles 76.2 cm	11.82	0.062	0.00525
80 m hurdles 76.2 cm	13.36	0.072	0.00536
80 m hurdles 84 cm	14.09	0.078	0.00550
100 m hurdles 76.2 cm	15.61	0.108	0.00693
100 m hurdles 84 cm	14.93	0.078	0.00523
100 m hurdles 91.4 cm	15.67	0.084	0.00534
110 m hurdles 91.4 cm	15.90	0.103	0.00646
110 m hurdles 100 cm	15.74	0.096	0.00612
110 m hurdles 106.7 cm	15.48	0.085	0.00548

Straight flat sprints cluster at 0.0059–0.0063 per m/s per second of race time; the bend events (150/200 m) sit about a third lower; hurdle codes bracket the flat rates.

Table S5: Quadratic specification: linear and quadratic wind terms (raw outcome units per m/s: seconds for sprints, metres for jumps, e.g. 0.0287 m = 2.87 cm; SEs clustered on meet).

Event	Linear term (SE)	Quadratic term (SE)	n
60 m	−0.0595 (0.0008)	0.0037 (0.0003)	72 669
80 m	−0.0752 (0.0048)	0.0013 (0.0019)	1 410
100 m	−0.0856 (0.0013)	0.0057 (0.0005)	64 756
150 m	−0.0794 (0.0141)	−0.0052 (0.0079)	334
200 m	−0.1306 (0.0033)	0.0141 (0.0011)	59 416
Long jump	0.0287 (0.0016)	0.0001 (0.0006)	51 870
Long jump (zone)	0.0194 (0.0010)	0.0015 (0.0004)	44 336
Triple jump	0.0372 (0.0034)	0.0020 (0.0011)	15 071
Triple jump (zone)	0.0342 (0.0048)	0.0030 (0.0014)	5 426

Negative linear terms for sprints mean faster times with tailwind. Positive quadratic terms in the large sprints match the headwind-tailwind asymmetry of Table 4.